BabyCare System

Step-by-Step Instructions to Replicate the Demonstrator

Team BabyCare – IEEE AP-S Student Design Contest 2026

1 System Overview

BabyCare is a compact automotive demonstration system designed to detect whether a baby is present in a car seat and to trigger a Bluetooth alert. The system combines RFID-based seat and occupancy detection with a continuous-wave radar used to identify breathing-related micro-movements. Figure 1 summarizes the functional architecture of the demonstrator, while Fig. 2 shows the corresponding physical implementation.
The BabyCare GitHub repository provides access to project resources, design files, and documentation: `https://github.com/M-guillaume/BabyCare`

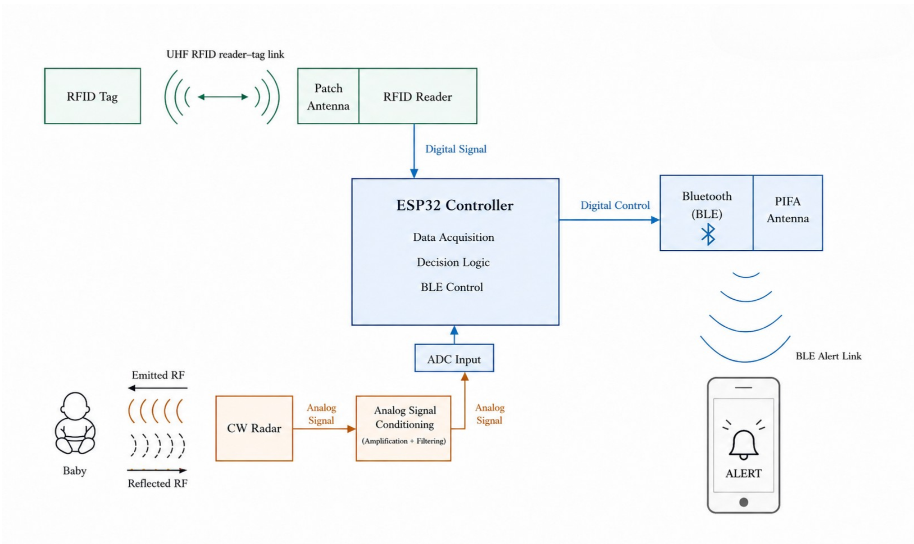

Figure 1: Functional architecture of the BabyCare demonstrator.

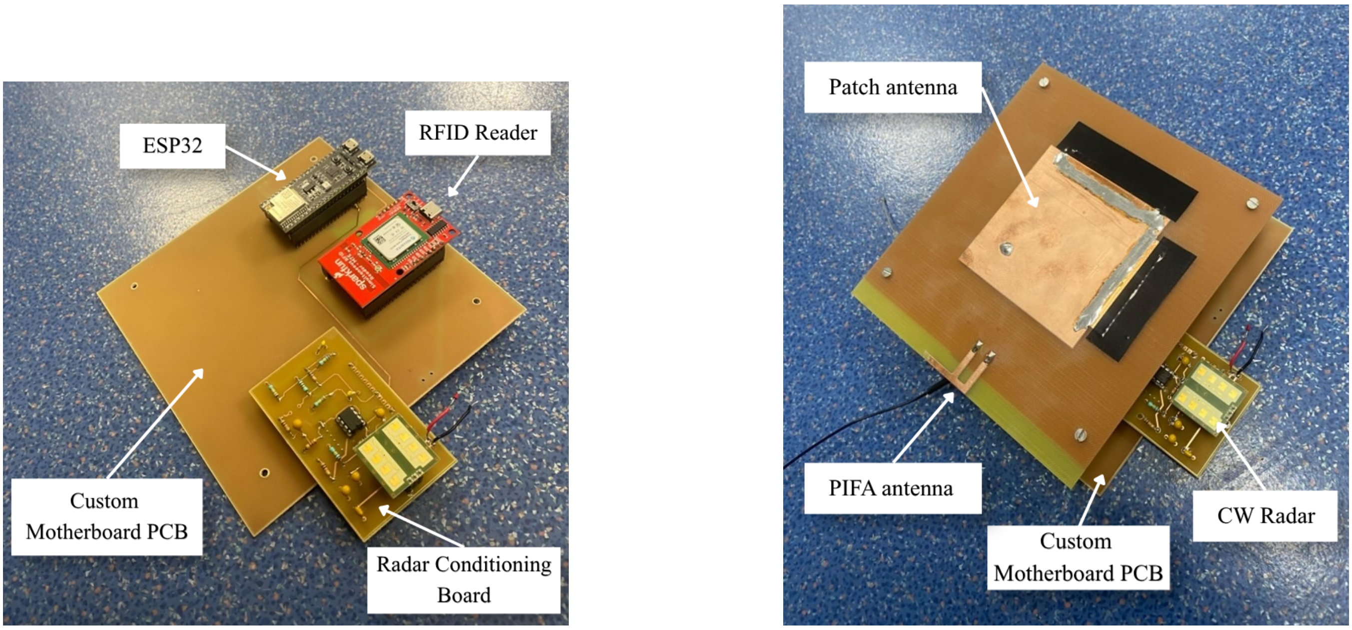

Figure 2: Hardware implementation of the BabyCare demonstrator: open assembly and complete assembly.

2 Required Components, Materials, and Tools

This section lists the main elements required to reproduce the BabyCare demonstrator. Detailed supplier references, costs, and purchasing links are provided in the final report. Equivalent components may be used if they satisfy the same electrical and mechanical requirements.

2.1 Electronic Components

Item	Qty.	Purpose / Requirement
ESP32-S3 board with antenna port	1	Main controller and Bluetooth communication
Bluetooth antenna	1	Wireless alert transmission
SparkFun Simultaneous RFID Reader – M7E Hecto	1	RFID tag detection
High-frequency RFID tags	2 min.	Seat and occupancy detection
IPM-165 CW radar module	1	Breathing-related motion detection
Operational amplifiers	TBD	Radar amplification stage
Resistors	TBD	Radar amplification circuit
Capacitors	TBD	Radar amplification circuit
Op-amp sockets / mounting parts	TBD	Assembly of the amplification board
20-pin header strips	6	Board interconnections and modular assembly
Female antenna connectors, solderable	2	Antenna-to-board connection
USB-C solderable connector	1	Power and/or programming interface
USB-C to USB cable	1	ESP32 programming and power supply
Infant breathing simulation plush	1	For system testing

Table 1: Electronic components required for the BabyCare demonstrator.

2.2 RF and Mechanical Materials

Item	Qty.	Purpose / Requirement
Copper-clad board, 16 × 16 cm	1	Antenna or RF board fabrication
Copper-clad board, size TBD	1	Additional antenna or circuit fabrication
U.FL to SMA male cable	1	RF connection between board and antenna
W.FL to SMA male cable	1	RF connection between board and antenna
Spacers	4	Mechanical positioning of boards or antennas
PMA plate, minimum size TBD	1	Mechanical support for the demonstrator
Connection wires / jumper wires	As needed	Electrical interconnections
Baby car seat	1	Demonstration support

Table 2: RF and mechanical materials required for the demonstrator.

2.3 Tools and Laboratory Equipment

Tool / Equipment	Purpose	Notes
Computer	Firmware upload and application setup	Arduino IDE or PlatformIO
Soldering station	Component and connector assembly	Required for PCB assembly
PCB milling/printing machine	PCB fabrication from layout files	Can be replaced by an external PCB manufacturer
Multimeter	Electrical continuity and voltage checks	Recommended
Smartphone	Mobile application and Bluetooth alert display	Android/iOS depending on the application

Table 3: Tools and equipment required to fabricate, assemble, and test the system.

3 Hardware Assembly

This section describes the hardware assembly procedure for reproducing the BabyCare demonstrator.

3.1 Download the Fabrication Files

1. Open the BabyCare GitHub repository.
2. Download the PCB layout file named `Motherboard_Layout`.
3. Download the PCB layout file named `Radar_Amplification_PCB`.

4. Download the antenna layout files for the RFID patch antenna and the ESP32-S3 PIFA antenna.
5. Download the enclosure files for the PMMA box.

3.2 Fabricate the Motherboard and Radar Amplification PCB

1. Use the downloaded PCB layout files to fabricate the motherboard and the radar amplification PCB.
2. The boards can be fabricated using a PCB milling/printing machine or ordered from an external PCB manufacturer.
3. After fabrication, visually inspect both boards to check for short circuits, incomplete traces, or damaged pads.

3.3 Solder the Electronic Components

1. Solder the required components on the motherboard by following the corresponding layout and bill of materials.
2. Solder the operational amplifiers, resistors, capacitors, headers, and connectors on the radar amplification PCB.
3. Pay attention to the orientation of polarized components and integrated circuits.
4. After soldering, check the continuity of the main power and ground lines with a multimeter.

3.4 Mount the ESP32, RFID Reader, Radar Board, and CW Radar

1. Plug the ESP32-S3 board into the dedicated headers on the motherboard.
2. Plug the SparkFun Simultaneous RFID Reader – M7E Hecto into the corresponding position on the motherboard.
3. Plug the radar amplification PCB into the motherboard.
4. Connect the IPM-165 CW radar module to the radar amplification PCB.
5. **Important:** when handling the CW radar module, wear an electrostatic discharge wrist strap and avoid touching the small antennas on the module. Electrostatic discharge or direct contact with the antennas may permanently damage the radar.
6. Check the orientation of the radar before plugging it in. The radar antennas must face the inside of the board, toward the component-free area reserved for the radar module.

3.5 Fabricate and Connect the Antennas

1. Use the antenna layout files from the GitHub repository to fabricate the RFID patch antenna and the ESP32-S3 PIFA antenna on copper-clad boards.

   Note: CST Microwave Studio simulation files for both antennas are also available in the GitHub repository. These files can be opened and modified using the free CST student license, making them suitable for further exploration or design optimization in an academic setting.

2. The RFID patch antenna is designed to operate at **915 MHz**.
3. The ESP32-S3 PIFA antenna is designed to operate at **2.45 GHz**.
4. For the patch antenna, solder the female antenna connector on the diagonal feed point to obtain the required 45°polarization and thus allow reading horizontally and vertically polarized tags.
5. For the PIFA antenna, solder the female antenna connector at the bottom of the F-shaped radiating structure.
6. Connect the RFID patch antenna to the RFID reader using the corresponding RF cable. In the BabyCare demonstrator, this connection is identified in red.
7. Connect the PIFA antenna to the ESP32-S3 antenna port using the corresponding RF cable.

3.6 Mechanical Mounting

1. Insert the four spacers at the four corners of the radar board.
2. Screw the radar board onto the motherboard using the spacers.
3. Make sure the radar module remains mechanically stable and correctly oriented.
4. Check that no metallic or conductive part creates an unwanted short circuit between boards.

3.7 Upload the ESP32 Firmware

1. Connect the ESP32-S3 to the computer using a USB cable.
2. Open the firmware provided in the BabyCare GitHub repository.
3. Select the correct ESP32-S3 board and serial port in the programming environment.
4. Install the required software libraries.
5. Upload the firmware to the ESP32-S3.
6. Open the serial monitor and check that the ESP32-S3, RFID reader, radar acquisition chain, and Bluetooth interface are correctly initialized.

3.8 Install the System in the Enclosure

1. Fabricate the PMMA enclosure using the laser-cutting files provided in the GitHub repository.
2. The recommended enclosure size is approximately 18×18 cm.
3. Place the assembled BabyCare electronic system inside the enclosure.
4. Add the required external connectors for system power and programming, including the USB-C access connector.
5. Make sure the RF cables are not excessively bent or mechanically stressed.
6. Close the enclosure while keeping access to the power connector and any required debugging connector.

3.9 Place the RFID Tags on the Baby Seat

1. Place the first RFID tag (Tag 1) under the foam of the baby seat or directly on the baby seat structure. This tag is used to detect whether the baby seat is installed. This tag is horizontally polarized.
2. Place the second RFID tag (Tag 2) on the side of the baby seat. This tag is used for the occupancy detection logic. This tag is vertically polarized.
3. Adjust the tag positions so that the first tag is reliably detected when the seat is present, while the second tag becomes blocked when the baby is seated.

3.10 Install the Mobile Application

1. Download the BabyCare mobile application from the GitHub repository.
2. Install the application on a smartphone.
3. Enable Bluetooth on the smartphone.
4. Pair or connect the smartphone to the ESP32-S3.
5. Verify that the application receives the BabyCare system state or alert message.

4 Demonstration Procedure

Test 1: Empty Car

Remove the baby car seat from the setup. The system should not detect Tag 1 and should remain in the initial state.

Test 2: Baby Seat Installed

Install the baby car seat with Tag 1 visible to the RFID reader. The system should detect the seat.

Test 3: Empty Seat

Keep the seat installed but leave it empty. Tag 2 should remain readable, indicating that the seat is not occupied.

Test 4: Occupied Seat

Place the baby mannequin or equivalent test object in the seat. Tag 2 should become blocked, indicating occupancy.

Test 5: Living Presence Detection

Activate the breathing demonstration. The CW radar should detect motion consistent with the simulated breathing activity. If motion is detected, the system confirms baby presence.

Test 6: Bluetooth Alert

Once the seat is detected, occupancy is confirmed, and breathing is detected, the ESP32 sends a Bluetooth alert to the smartphone.

5 Expected System States

State	Tag 1	Tag 2	Radar / Output
No seat	Not detected	–	No alert
Seat empty	Detected	Detected	No alert
Seat occupied by object	Detected	Blocked	No breathing, no alert
Baby detected	Detected	Blocked	Breathing detected, alert sent

Table 4: Expected behavior of the BabyCare demonstrator.

6 Troubleshooting

- If Tag 1 is not detected, reduce the distance between the RFID reader and the seat tag.
- If Tag 2 is always detected, adjust its position so that the body blocks the RFID path.
- If Tag 2 is never detected, check the tag orientation and RFID reader range.
- If no breathing signal is detected, realign the radar toward the chest area and reduce mechanical vibrations.
- If the Bluetooth alert is not received, check the ESP32 pairing status and restart the mobile application.

7 Reproducibility Notes

The setup should be reproduced using the same antenna positions, tag placement, firmware version, and test sequence described above. Minor adjustments of distances and orientations may be required depending on the RFID reader range, antenna geometry, and radar sensitivity.